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LedgerIQ

AI Finance Controller — Close the loop between orders, payments and settlements.

An illustration of a person in a business suit sitting at a desk, looking overwhelmed. The desk is cluttered with multiple computer monitors displaying various data, stacks of papers, and a calculator. The person is holding a pen and looking down at a document. The background is a light blue gradient.

The Merchant Finance Problem

Fragmented Sources

Orders, gateway, bank settlements arrive independently – never in sync

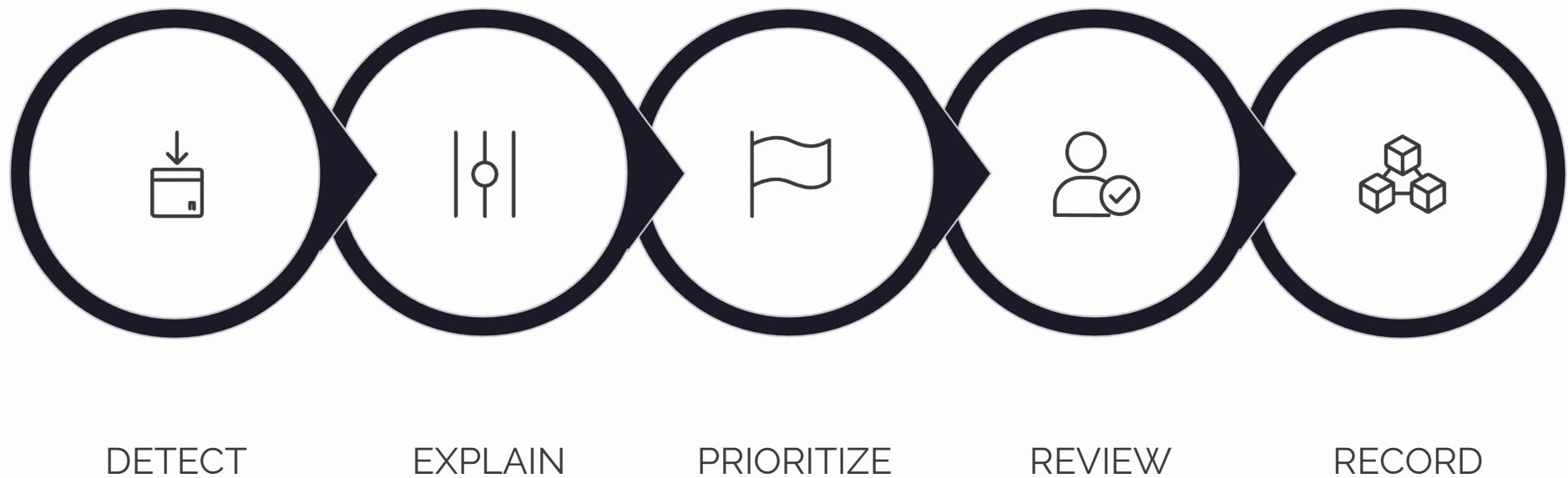
Manual Reconciliation

Slow, error-prone, hard to audit and impossible to prioritize at scale

Financial Exposure

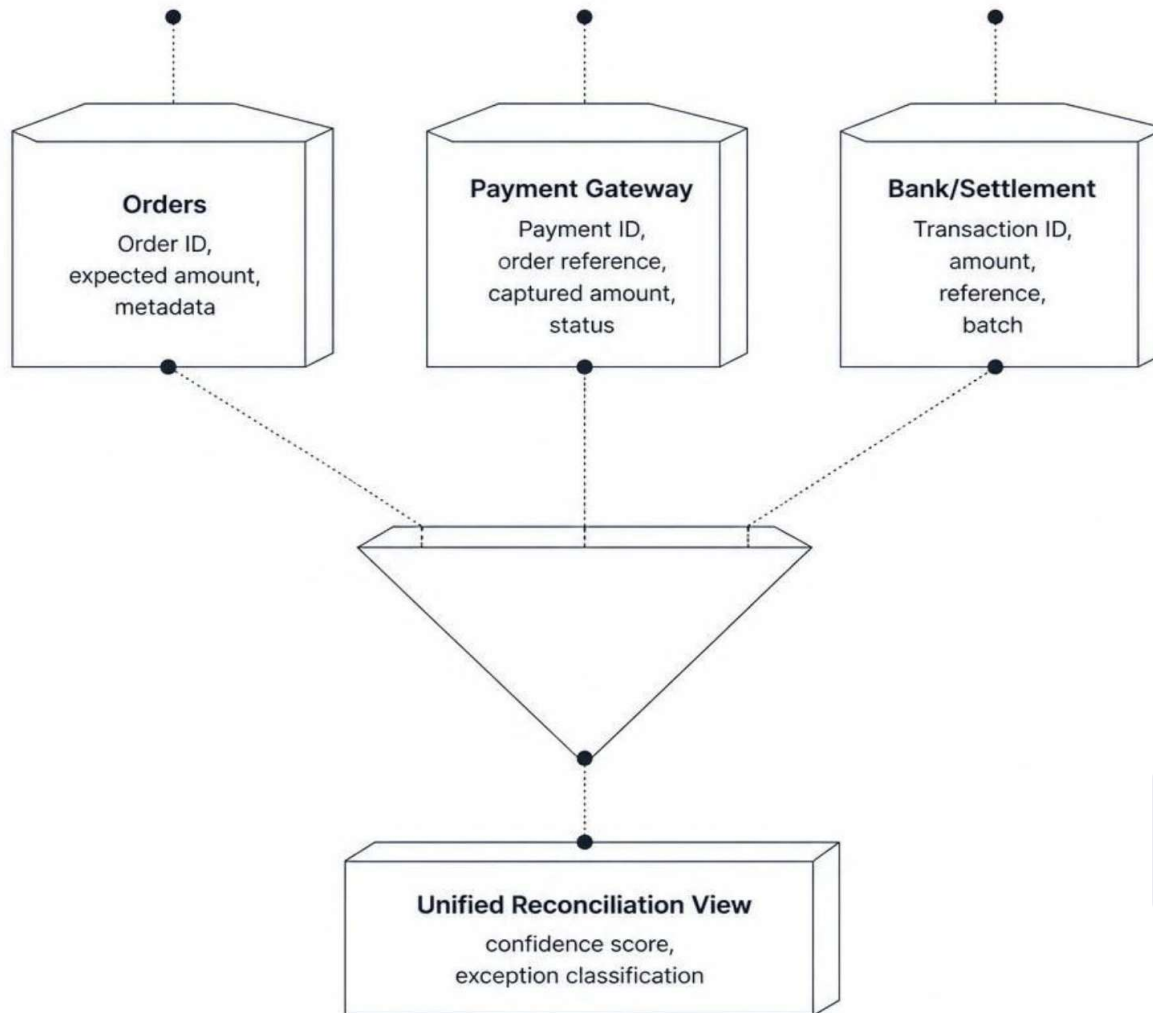
Unresolved mismatches create **amount-at-risk** that finance teams cannot see

The Finance Control Loop



Every merchant transaction travels through a closed, auditable loop – nothing falls through the cracks.

Data Reconciliation Pipeline



Three Sources. One Truth.

- **Orders:** Order ID, expected amount, metadata
- **Payment Gateway:** Payment ID, order reference, captured amount, status
- **Bank / Settlement:** Transaction ID, amount, reference, batch

Output: Unified reconciliation view with confidence scoring and exception classification

Deterministic Reconciliation Engine

1

Identifier Matching

Link orders → payments → settlements using order references

2

Amount Comparison

Detect mismatches and partial captures with precision

3

State Validation

Verify payment and settlement status consistency

4

Confidence Scoring

Calculate match probability based on evidence strength

HIGH CONFIDENCE

Matched – auto-recorded

MEDIUM

Flagged for human review

LOW

Exception – investigate

Exception Intelligence & Amount-at-Risk

Exception Classification

Probable Match

Near-match, needs confirmation

Gateway-Only

Payment captured, no settlement

Bank-Only

NOT silently matched — flagged as financial exposure

Amount Mismatch

Captured $\neq$ expected amount

Conservative Design Principle

Bank-only transactions are **never silently matched**. They surface as explicit financial exposure — because trustworthy reconciliation means showing what you don't know.



Amount-at-risk aggregates all unresolved exception values so finance teams always know where money is unaccounted for.

AI Judgment: Deterministic + Reasoning

Deterministic Core

- Identifier matching
- Amount comparison
- Reconciliation rules
- Confidence calculation
- Metrics and reporting

Reproducible. Auditable. Always correct.

Finance Copilot (AI Layer)

- Natural-language questions
- Reconciliation explanations
- Investigation assistance
- Recommended actions

AI assists reasoning – **deterministic logic protects financial correctness.**

Human Review & Audit Trail

01

Investigate

Finance team opens exception, reviews system explanation and recommended action

02

Decide

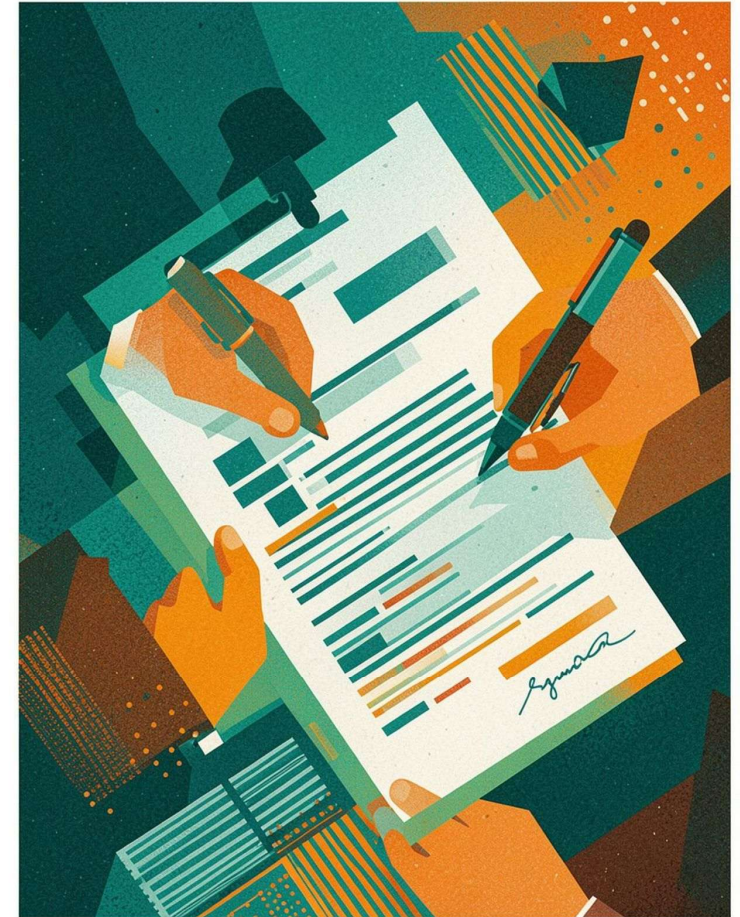
Approve, reject, or request more information — **human remains in control**

03

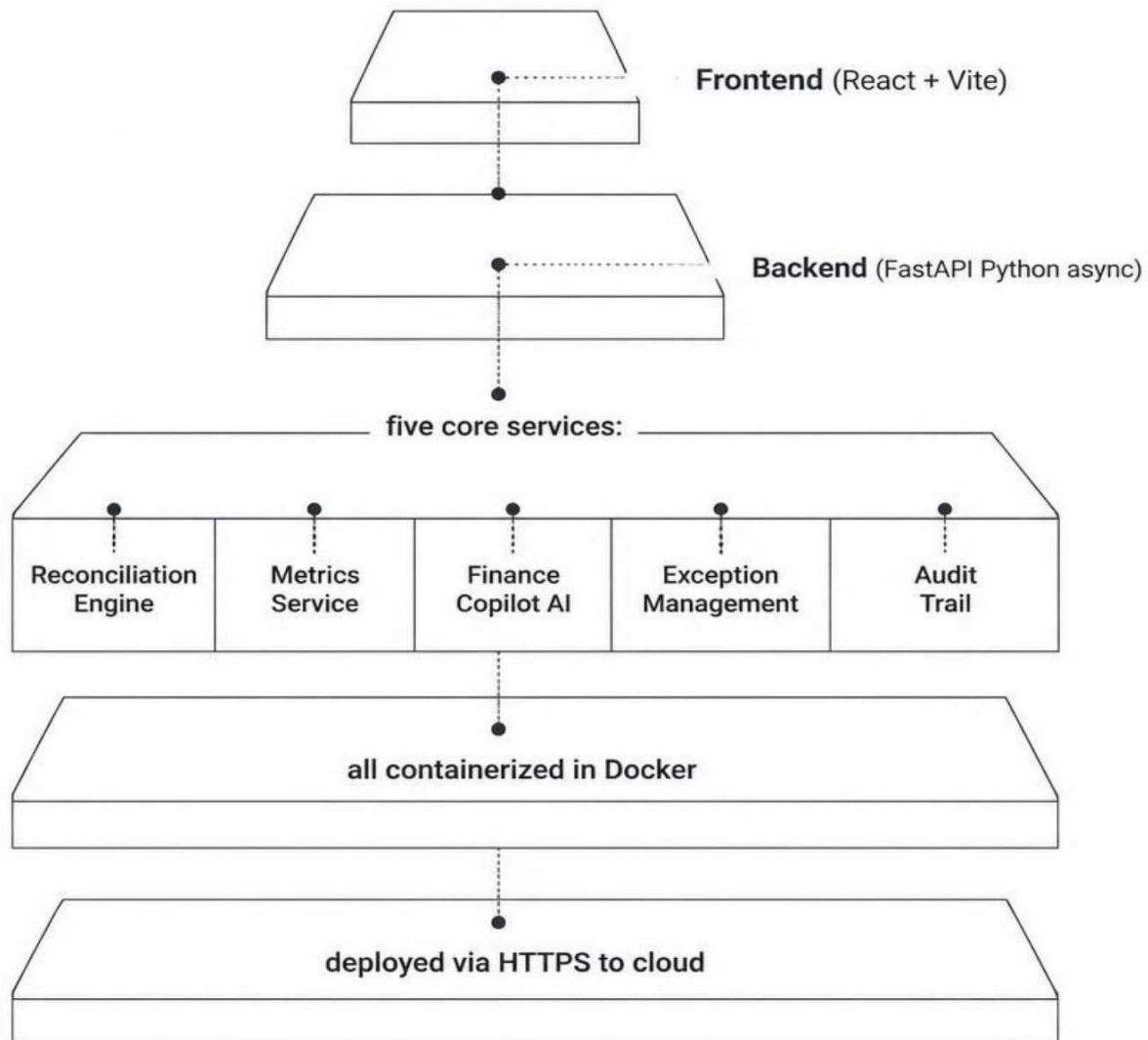
Record

Every reconciliation, query, review and decision is timestamped and traceable

"Every important operational decision is auditable."



Technical Architecture



Stack

Frontend

React + Vite — responsive, fast

Backend

FastAPI (Python, async, production-ready)

Deployment

Docker containerized · HTTPS hosted · Cloud-ready

Core Services

Reconciliation engine · Metrics · Finance Copilot · Exception management · Audit trail

Why LedgerIQ is Buildathon-Worthy

Real Problem

Merchant finance teams manually reconcile thousands of records daily — this is a live pain point, not a theoretical exercise

Trustworthy Design

Deterministic matching + explicit confidence + human review + immutable audit trail

Technical Depth

Data validation, normalization, decision logic, metrics, AI reasoning, production deployment

The Differentiator

Not just a dashboard — a **complete finance control loop** that keeps humans in control while automating the obvious and surfacing the risky